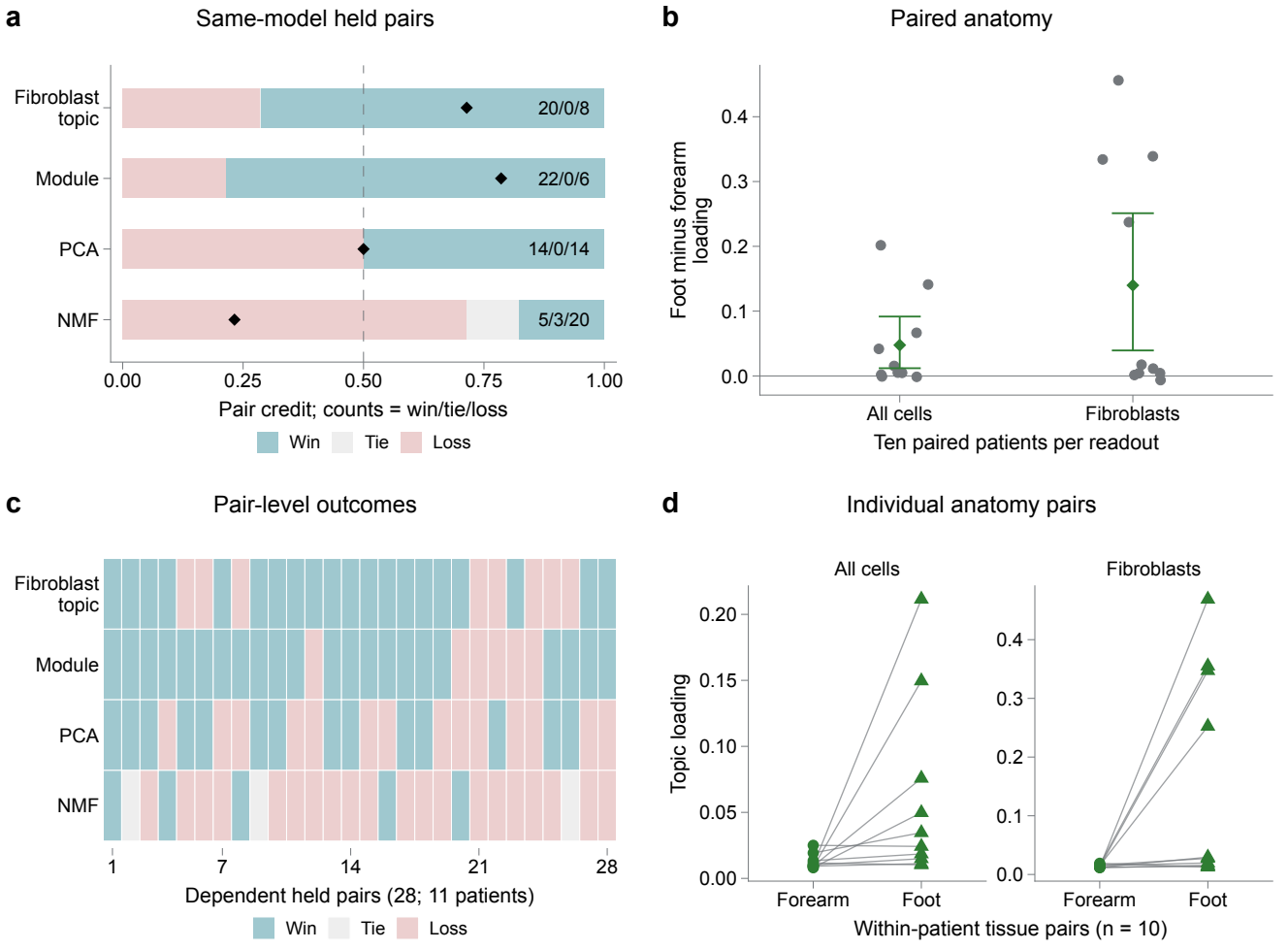
a
Same-model held pairs
Fibroblast topic
Module
PCA
NMF
20/0/8
22/0/6
14/0/14
5/3/20
0.00
0.25
0.50
0.75
1.00
Pair credit; counts = win/tie/loss
Win
Tie
Loss
b
Paired anatomy
Foot minus forearm loading
0.0
0.1
0.2
0.3
0.4
All cells
Fibroblasts
Ten paired patients per readout
c
Pair-level outcomes
Fibroblast topic
Module
PCA
NMF
1
7
14
21
28
Dependent held pairs (28; 11 patients)
Win
Tie
Loss
d
Individual anatomy pairs
All cells
Fibroblasts
Topic loading
0.00
0.05
0.10
0.15
0.20
0.0
0.1
0.2
0.3
0.4
Forearm
Foot
Forearm
Foot
Within-patient tissue pairs (n = 10)